Cross-Lingual Citation Verification Using Multilingual Transformer Models

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Abstract

This paper addresses the problem of cross-lingual citation verification in academic writing. The problem arises when a thesis cites a source written in another language and then reuses its content through translation or paraphrase, so the citation may look formally correct even though the source does not clearly support the claim. Standard plagiarism detection systems are not designed for this task because they focus mainly on textual overlap rather than semantic support between a citing statement and its cited source. The main result is an end-to-end verification pipeline that takes a PDF thesis, extracts numerical style in-text citations and bibliography entries, resolves source metadata through DOI-based lookup, retrieves full text or abstracts when available, segments the retrieved documents into passages, and ranks those passages against the citing context by using multilingual transformer embeddings. The intended output is a protocol-style report that supports human review instead of replacing it with a single automatic verdict. The paper also reports an evaluation on 1000 Slovak bachelor and master theses from four faculties. The results show that full-text availability remains the main practical bottleneck. Even with that limitation, the pipeline can still separate likely supporting passages from weaker cases and reduce the amount of manual search required during citation review.

Index Terms

cross-lingual citation verification, multilingual embeddings, academic integrity, plagiarism detection, scholarly document analysis

I. Introduction

Academic writing depends on citations not only for attribution but also for verification. A reader should be able to follow a reference and confirm that the cited source really supports the claim made in the text. Recent studies show that this link is not always reliable. A citation may be present, yet the cited source may support the claim only partially, be interpreted incorrectly, or be used in a misleading way [1] [2] [3].

This problem overlaps with plagiarism, but it is not identical to it. Plagiarism is usually associated with direct copying, but in practice the broader issue also includes misleading paraphrase, weak attribution, and reuse of ideas that is formally cited but still ethically problematic [4]. Citation misuse belongs to that wider category because a source can be acknowledged and still fail to justify the statement to which it is attached.

The difficulty becomes greater in cross-lingual writing. A Slovak thesis may cite an English article, translate the relevant idea, compress it into one sentence, and provide a formally correct reference. In that case, the central question is not whether the source was cited, but whether it actually supports the translated claim. This places the task between cross-lingual plagiarism detection, which studies reuse across languages [5], and citation verification, which asks whether the cited source supports the citing statement [6].

This matters in real review practice. Systems such as Turnitin [7] and CRZP [8] are useful when the goal is to compare suspicious text against large repositories and flag lexical overlap. They are not designed to guide a reviewer from one citation context in a thesis to the most relevant supporting passage in a foreign-language source. In practice, that check is usually manual. The reviewer must find the source, locate the relevant passage, and decide whether the claim is justified. That takes time and does not scale well.

The intention of this system is not to build a fully automatic misconduct detector, but a reviewer-support tool that makes the evidence trail easier to inspect. Instead of treating citation verification as one model decision, it treats the task as a pipeline that begins with a noisy PDF file and moves through citation extraction, reference matching, source retrieval, passage selection, and semantic comparison. Each step can fail in a different way, so the system should keep uncertainty visible instead of hiding it behind one score.

This article has four main contributions. First, it frames cross-lingual citation verification as an evidence retrieval and ranking task rather than a standard plagiarism score. Second, it proposes an end-to-end pipeline that links local citation contexts with multilingual source passages. Third, it reports results from 1000 Slovak bachelor and master theses collected from four university faculties through CRZP. Fourth, it discusses practical limits such as source access, threshold setting, and reviewer workflow.

II. Related Work

A. From plagiarism detection to citation verification

The closest neighbouring area is plagiarism detection. Reviews of the field describe a gradual move from simple text matching toward more semantic and structural approaches, mainly because reused text is often changed before it appears in the suspicious document [4]. That trend is relevant here, but citation verification asks a narrower question. The goal is not only to detect reuse, but to decide whether a cited source provides evidence for a specific claim.

In cross-lingual settings, this becomes harder because the source and the suspicious text are written in different languages. Early approaches relied on dictionary-based [9] translation or syntax-based [10] comparison. Classical lexical methods such as string matching [11] and n-gram [12] comparison can still work when the suspicious text stays close to the source. Their weakness is that performance drops when the author translates, paraphrases, or compresses the original meaning more aggressively [13]. Amirzhanov et al. [13] describe cross-lingual plagiarism detection as a problem that increasingly requires multilingual rather than purely lexical comparison.

Citation verification is related, but it asks a different question. Plagiarism detection asks whether material was reused and where it came from. Citation verification asks whether the cited source really supports the statement in the thesis. That distinction matters because a citation can be formally correct and still be inaccurate, incomplete, or misleading [1] [2].

B. Multilingual semantic representations

The shift toward semantic comparison became much more practical with multilingual neural models. BERT [14] showed that contextual pretraining can learn richer text representations than simple lexical features. XLM-R [15] extended this idea to many languages and remains a strong baseline for cross-lingual semantic tasks. Newer encoder families such as XLM-V [16] continue this line by improving multilingual coverage and representation quality.

This is directly relevant to citation verification because the system compares short citing sentences with short candidate evidence passages. Those units need to be placed in one shared semantic space, and exact wording overlap is often too weak a signal in multilingual academic writing. Recent studies report good results for transformer-based multilingual embeddings in paraphrase-oriented and semantic cross-lingual settings [17].

C. Citation analysis

Some citation research focuses on rhetorical role rather than factual support. SciCite, for example, classifies citation intent in scientific writing [18]. This is useful background because not every citation serves the same argumentative function. Some provide context, some compare methods, and some are meant to support a concrete claim.

Work closer to the goal of this paper asks whether the source actually supports the citing statement. Mogull [1] separates minor citation inaccuracies from major content errors, where the cited source fails to substantiate or even contradicts the assertion. Pavlovic et al. [2] show that such claim-support failures also occur in real published literature. Mehregan [3] further notes that some unsupported citations may reflect deliberate manipulation rather than simple carelessness.

Zeeb and Olsson explicitly study whether large language models can evaluate citations in scholarly documents [6]. Their setup treats citation evaluation as a binary decision with a short explanation. Haan's SemanticCite is even closer at the systems level because it approaches citation verification as a workflow over full texts, citation contexts, and evidence passages [19]. These works are important because they show that the task can be framed as evidence retrieval and reasoning, not only as document-level similarity.

D. Institutional systems and the remaining gap

Institutional systems still matter because they define how plagiarism checking works in practice. Turnitin [7] and iThenticate [20] are widely used in broader academic workflows, while Odevzdej [21] and CRZP [8] play a similar role in Czech and Slovak environments. Their strength is large-scale indexing and practical reporting. Their limitation is that they are designed primarily for monolingual plagiarism detection, not for multilingual citation verification.

Recent work also warns that evaluation choices can distort the apparent reliability of automated systems. Threshold selection, class imbalance, and overconfident interpretation can make reported performance look cleaner than the underlying evidence really is [22] [23]. Any reviewer-support system should keep uncertainty visible instead of presenting heuristic similarity scores as final misconduct decisions.

That is the gap this work addresses. It does not try to replace large plagiarism platforms or reproduce their repository scale. Its goal is narrower: to help reviewers with more efficient evidence retrieval and ranking in cross-lingual citation verification process.

III. System Design

The proposed solution is built as a pipeline, not as one single classifier. That choice is practical. Cross-lingual citation verification works with several different kinds of data: PDF pages, bibliography strings, document metadata, source text, short passages, and semantic vectors. Each stage brings its own errors and uncertainty. A pipeline makes those steps visible, easier to debug and more maintainable in the future.

A. Design principles

Three ideas guided the design. The first is locality. Citation support is usually a local relation between one claim and one or a few source passages, so the system should not work only at document level. The second is graceful degradation. If full text is missing, the system should still keep metadata and use an abstract when possible. The third is human-centered output. The tool is meant to save time for reviewers, not to replace them.

B. Document understanding and reference alignment

The input is an academic PDF with numeric citations. The system first extracts the in-text citation markers and the bibliography section. The restriction to numeric citations is deliberate. Square-bracket references are easier to detect reliably than author-year styles, footnote citations, or mixed local formats. This keeps the work focused on verification instead of spending most of the effort on citation parsing.

Each bibliography entry is converted into a structured record with the raw reference string, DOI, the citing sentences, and how often the source is cited in the thesis. The system then links each local citation context to one reference item. That citation-reference pair is the basic unit used in the later stages.

PDF extraction is one of the weak points of the whole process. Broken line wraps, OCR noise, and inconsistent punctuation can all make citation parsing less reliable. Because of that, the system should not pretend that the first stage is exact. It needs to preserve uncertainty instead of hiding it.

C. Source resolution and retrieval

After the system links a citation to a reference, it tries to resolve that reference to a real source document. Retrieval follows a layered strategy. If the reference contains a DOI, the system uses it in a Crossref [24] API request to obtain precise metadata. If no DOI is available, it falls back to Crossref full-text search. If neither route returns a reliable match, the source is marked as unavailable.

When a probable metadata match is found, the next problem is source access. The system first checks whether the full text can be obtained through one of the publisher APIs available to it. Many publishers protect their websites against automated scraping, so an official API is often more reliable than direct HTTP requests. If no API route is available, the system falls back to standard HTTP retrieval. When the full text still cannot be obtained, it tries to retrieve at least the abstract. If no usable source text is found, the reference remains unavailable in the final report.

This layered approach is necessary because academic sources vary widely in accessibility. Some are openly downloadable, some expose only metadata or an abstract, and others are blocked by publisher restrictions. Retrieval failure is therefore informative in its own right and should remain visible in the output instead of being hidden.

After a source has been obtained, the text is converted into machine-readable structure and segmented into smaller passages with GROBID [25]. This matters because whole-document similarity is usually too coarse. A paper may address the right topic overall, while only one specific passage directly supports the thesis statement. Passage-level comparison is therefore more informative than document-level matching.

D. Multilingual semantic ranking

The semantic stage maps citing sentences and source passages into one multilingual embedding space. The system then compares them with cosine similarity. This is the part that makes cross-lingual verification possible. The thesis sentence and the source passage do not have to use the same language or the same wording, but they can still be compared through their vector representations.

The testing used an XLM-R-based [15] sentence embedding model because it offers broad language coverage and good practical performance in cross-lingual similarity tasks [17]. The design is not tied to one model, though. Other multilingual encoders could also be used, if they produce suitable sentence-level embeddings.

The similarity score is not treated as proof that the citation is correct or incorrect. Instead, the system uses configurable thresholds to label the best match as *supported*, *related*, or *no_support*. These labels are meant for triage. They help the reviewer see which citations look strong and which ones need more attention.

E. Protocol-style output

The final output is a protocol-style PDF report. For each processed reference, the report can list citation frequency, extracted citation contexts, retrieval status, identifiers, and the passages that scored best. This format is more useful than one number because citation problems do not all look the same. Some come from parsing errors, some from inaccessible sources, and some from weak semantic support.

The system therefore automates the repetitive parts of the task, but it leaves the final interpretation to a human reviewer.

IV. Experimental Setup

A. Corpus construction

The main experiment used a corpus of 1000 bachelor and master theses collected from the Central Register of Theses and Dissertations [8]. The sample contains 250 documents from each of four faculties focused on computer science or software engineering. The theses were submitted between 2020 and 2025. In the result section, the faculties are labeled Faculty1 through Faculty4 because the point of the experiment is to study the method, not to rank institutions.

This corpus fits the research question well. Slovak technical theses often cite English-language papers, documentation, and conference articles. That creates a realistic setting for cross-lingual citation verification instead of an artificial benchmark built from manually prepared sentence pairs.

B. Execution environment

The experiments were run on a workstation with an AMD Ryzen 5 7600 CPU, 32 GB RAM, and an NVIDIA GeForce RTX 4070 Ti Super GPU. The GPU was used for multilingual embedding inference. PDF parsing, metadata handling, source retrieval, and storage ran on the CPU.

C. Reproducibility strategy

The thesis implementation was designed so that it could be run again under the same conditions. Dependency versions were pinned, structured text extractions were cached, and embeddings were stored for reuse. These choices reduce repeat cost and make the outputs more stable when the same documents are processed again with the same model.

Full end-to-end reproducibility is still limited by outside services. DOI resolution can change, publisher pages can behave differently over time, and some retrieval failures depend on network conditions. Because of that, source retrieval is not just a side issue. It is part of the real behavior of the system.

The experiment does not treat model output as a misconduct decision. The similarity thresholds were chosen as engineering settings for a prototype, not as formal academic or legal boundaries. The labels are meant to divide reviewer workload into likely support, possible relevance, and weak support. This matters because automated integrity tools can easily look more certain than they really are [22].

V. Results

A. Reference quality and DOI coverage

The corpus contains enough structured metadata to make large-scale source resolution feasible, but the quality is not uniform across faculties. Figure 1 shows that mean DOI coverage reaches 77.4% in Faculty 2, 70.2% in Faculty 4, and 60.1% in Faculty 3, while Faculty 1 is much lower at 42.8%. The error bars also show substantial variation between theses from the same faculty, so DOI availability is not only a faculty-level property.

The DOI is the main entry point for the retrieval pipeline. A thesis whose bibliography contains many references without a valid DOI immediately gives the system fewer sources that can be checked through metadata resolution and downstream download.

B. Source retrieval outcomes

Figure 2 shows the retrieval outcomes in the final experiment. Across the broader run, 20 631 references were associated with a DOI, 7 175 source documents were successfully downloaded, and 2 551 additional abstracts were retrieved and verified. That corresponds to roughly 34% of all DOI-associated references when only full-text documents are counted.

At faculty level, downloadable full-text PDFs account for 14.5% of references in Faculty 1, 29.8% in Faculty 2, 21.9% in Faculty 3, and 28.2% in Faculty 4. Abstract fallback adds another 5.4%–10.4%. Faculty 1 also has the largest *no_valid_doi* share, which is consistent with its weaker DOI coverage. In Faculties 2–4, the main obstacle is *download_failed*, which ranges from 35.6% to 42.1%.

The final experiment used Crossref metadata resolution together with arXiv and Elsevier API access. IEEE and Springer integration had already been implemented, but access arrived too late for a full rerun before submission. If these two APIs were available, the download rate is expected to be around 60% of DOI-associated references, which would be a substantial improvement over the 34% achieved in the current run.

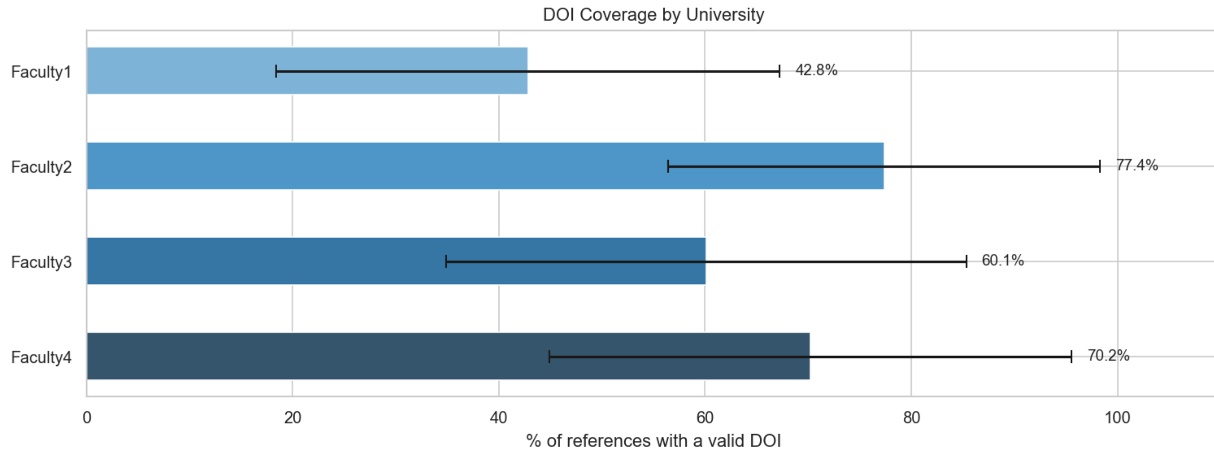

Fig. 1. Mean DOI coverage by faculty. Error bars show standard deviation within each faculty sample.

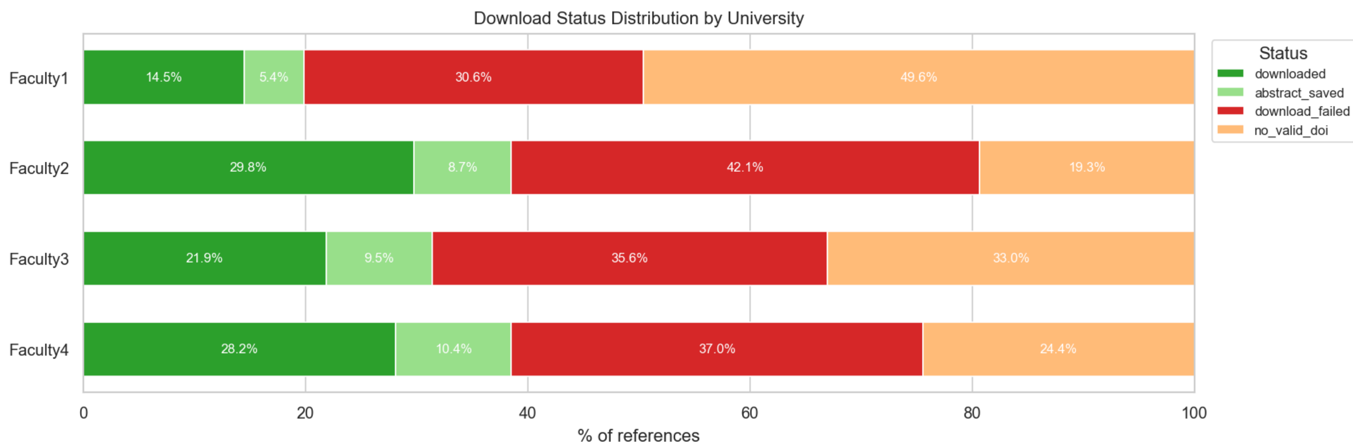

Fig. 2. Source retrieval outcomes by faculty.

Publisher-specific API integrations were prioritised based on the observed DOI distribution that can be seen in Figure 3. IEEE, Springer Nature, Elsevier, and arXiv together cover 47% of DOI-bearing references in this corpus, so dedicated handling for these four yields the greatest immediate gain. Beyond them the distribution is long-tailed, with many additional publishers each contributing only a small number of references. A corpus from a different discipline would produce a different publisher ranking and therefore a different optimal set of integrations.

C. Similarity structure

Figure 4 shows the distribution of best cosine similarity scores across all verified citation contexts. The distribution is broad, with its main concentration in the intermediate region between roughly 0.45 and 0.65. Very low scores and very high scores do occur, but they are much less frequent than mid-range values.

This means the verified citation-source pairs do not split cleanly into two obvious groups. Many cases fall into a semantically related middle ground, which is consistent with the way citations are used in academic writing. Authors often summarize, paraphrase, or partially adapt source material instead of reusing one clearly matching sentence.

D. Support labels across faculties

When the similarity scores are mapped to the three working labels, the *related* category is the largest in every faculty, ranging from 43.3% in Faculty 1 to 50.7% in Faculty 3, as shown in Fig. 5. The share of citations marked as *supported* ranges from 16.4% in Faculty 1 to 26.2% in Faculty 4. The *no_support* category ranges from 24.0% in Faculty 2 to 40.3% in Faculty 1.

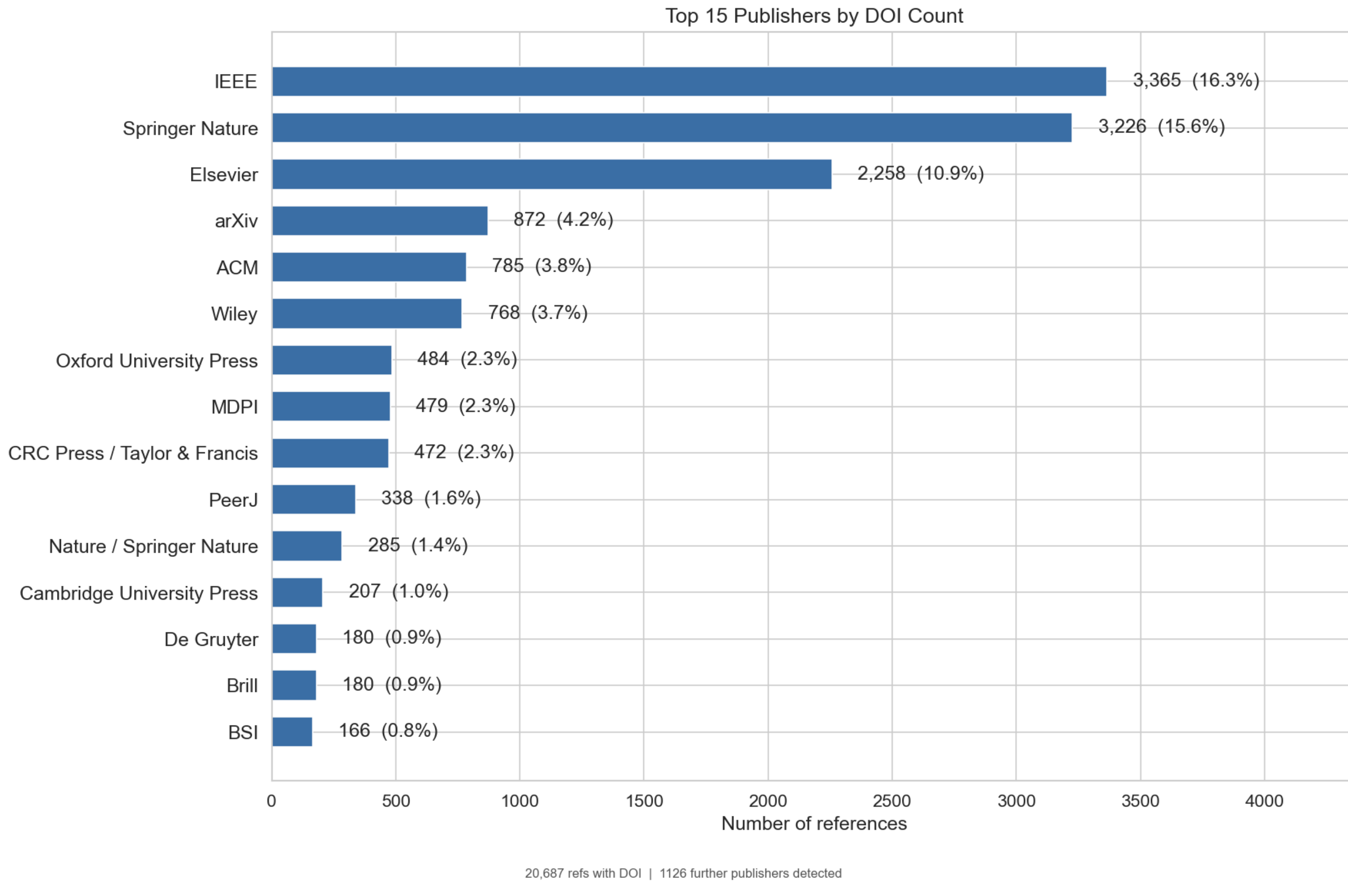

Fig. 3. Source retrieval outcomes by faculty.

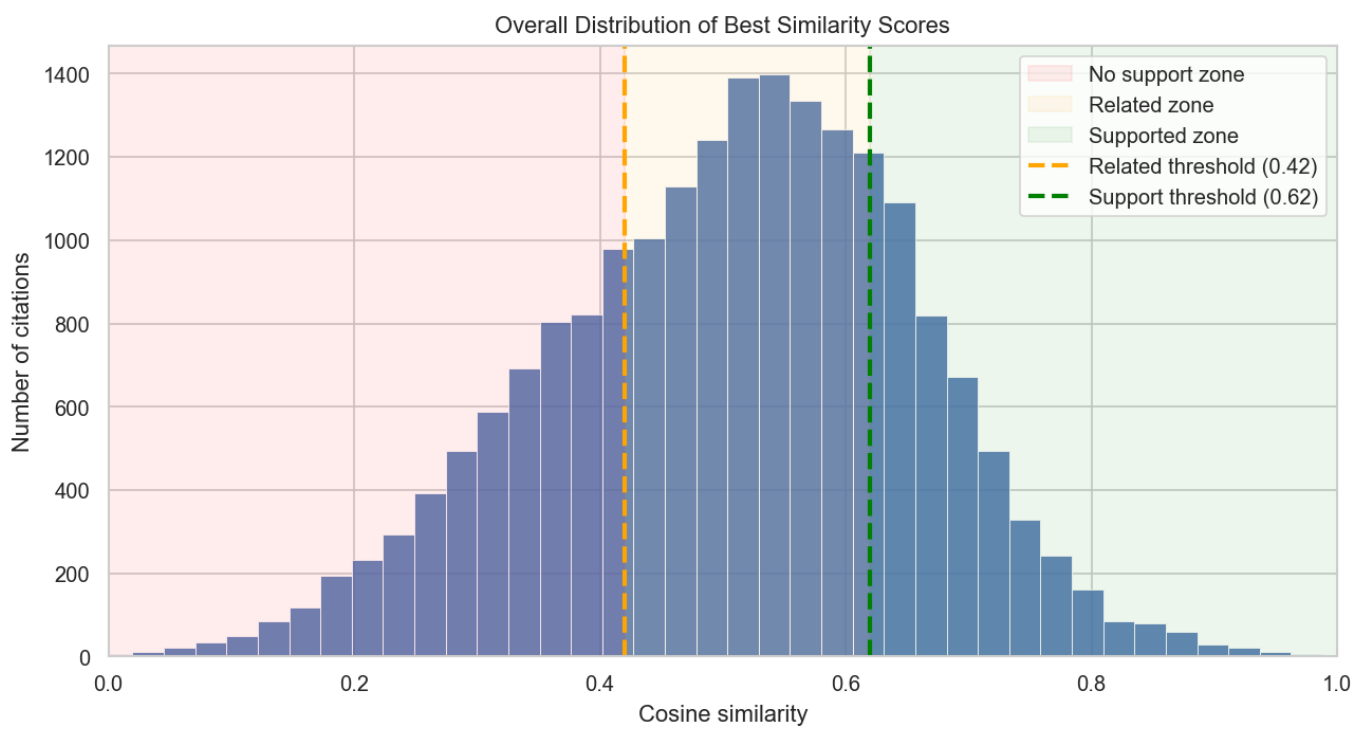

Fig. 4. Distribution of best similarity scores across verified citation contexts.

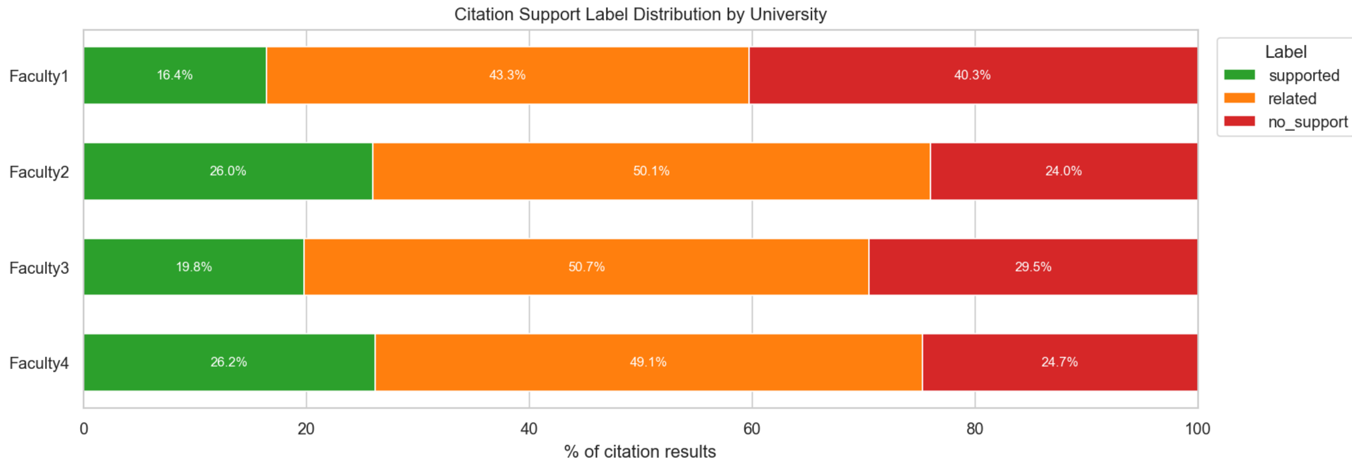

Fig. 5. Distribution of citation-support labels among verified citation contexts.

Faculty 1 stands out with the lowest supported share and the highest no-support share, but this should be interpreted carefully. The same faculty also has the weakest DOI coverage and the lowest retrieval success, so part of the difference reflects the amount and quality of evidence available to the pipeline rather than citation practice alone.

E. Citation reuse and verification workload

Figure 6 shows that the dominant pattern in every faculty is single use of a reference. The largest category is references cited exactly once, ranging from 44.7% in Faculty 1 to 65.3% in Faculty 4. References cited twice form the next largest group, while heavier reuse of four or more occurrences remains a smaller share at 8.9%–17.9%.

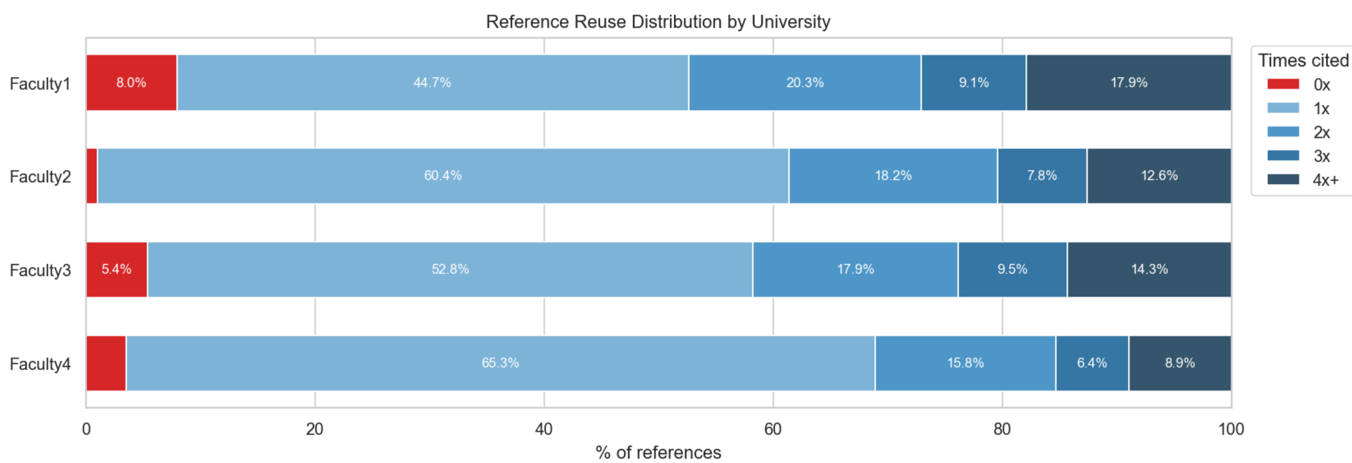

Fig. 6. Reference reuse distribution by faculty.

Most sources are used locally for one or a few claims, but a smaller subset of references appears repeatedly in different parts of the thesis. Repeated reuse makes the task more context-specific, because one source may serve several argumentative roles and therefore require different supporting passages for different citation contexts.

F. Meaning of low-support cases

The *no_support* label needs careful interpretation. A low cosine similarity between a citing sentence and the best retrieved passage does not by itself show that the citation is false or misleading. The student may be summarizing a broader argument, paraphrasing at a higher level of abstraction, or the system may have downloaded an incomplete or inaccurate version of the source.

That is why the labels should be read as reviewer-support signals rather than final decisions. High-similarity cases can often be checked quickly, while lower-similarity cases deserve closer manual reading. The pipeline narrows the search space and makes evidence easier to inspect, but it does not replace human judgment.

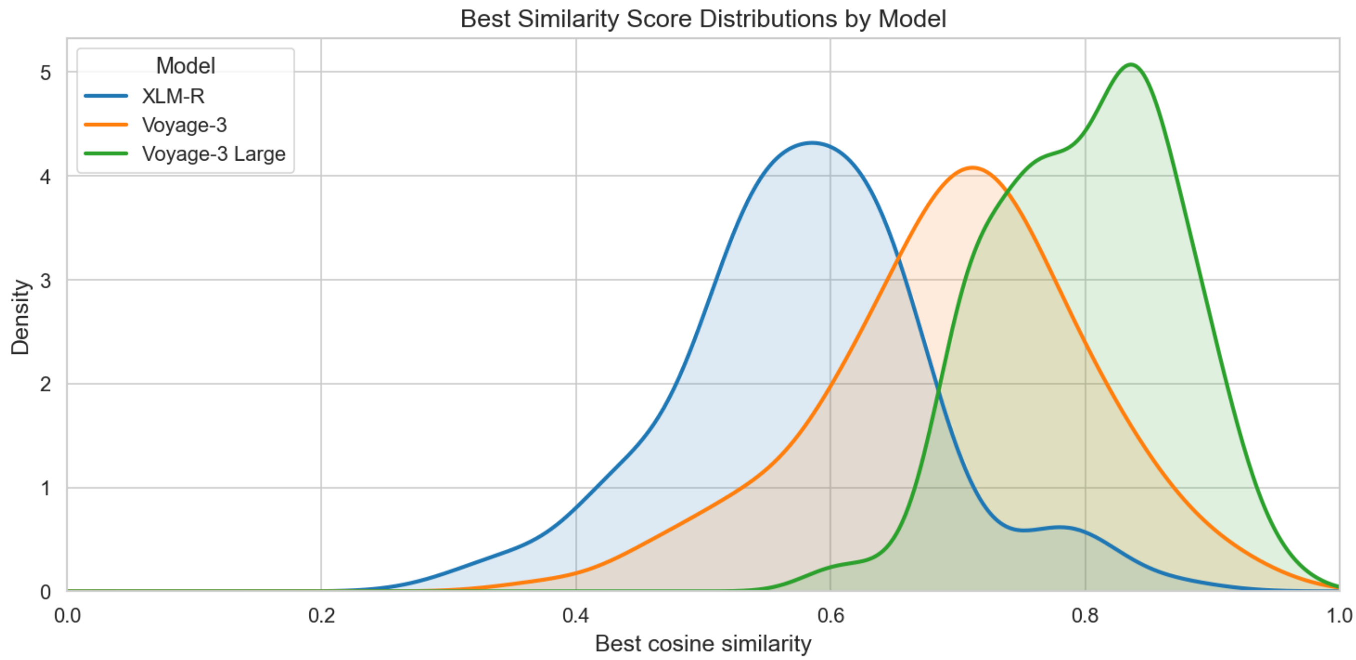

Fig. 7. Similarity-score distributions for the same cited sentences across XLM-R, Voyage 3, and Voyage 3 Large embeddings.

VI. Discussion

A. Why the pipeline matters

One of the main lessons of the project is that cross-lingual citation verification is not only a modeling problem. The final result depends on a chain of earlier steps: PDF parsing, bibliography extraction, DOI detection, source access, text segmentation, and only then semantic comparison. Because of that, it makes more sense to view the task as a workflow than as one classifier output.

This is useful from a software point of view as well. If one step needs to be improved later, its internal processing can be changed without causing problems in the rest of the system, as long as the same input and output are preserved. That means the parser, retrieval logic, or semantic model could be replaced in the future without redesigning the whole pipeline.

B. Validity limits

Threshold validity remains an open issue. The labels used in this paper are useful for triage, but they are not based on any manually annotated benchmark. A better next step would be to build a gold set of citation contexts, candidate source passages, and reviewer judgments, and then tune the thresholds against human decisions instead of heuristic expectations. It is important to note that this tuning depends on the embedding model. If a different model is used in the future, the embeddings will change, and the similarity scores will change with them. A small experiment was conducted to prove this. Figure 7 shows similarity-score distributions for more than 200 cited sentences when the same text pairs are evaluated with XLM-R, Voyage 3, and Voyage 3 Large embeddings. The distributions differ across models, even trough the same model family, which means that the same citation-source pair can receive a meaningfully different score depending on the embedding backend. That makes model-specific calibration necessary and means the current threshold values should not be treated as fixed constants for all future versions of the system.

Evidence validity is another limit. A citation can still be valid even when no short passage looks clearly supportive. Authors often compress several paragraphs into one sentence, rely on shared field knowledge, or paraphrase at a level of abstraction that sentence embeddings do not capture well. Translation adds another difficulty because idioms and terminology shifts can lower similarity even when the source is used correctly [26].

C. Deployment considerations

If the system is developed further, it should become more interactive rather than more automatic. A practical reviewer interface should let users inspect extracted bibliography entries, confirm, reject or add source matches, and browse the selected evidence passages before a report is finalized. That would make the tool more robust when the input PDF is noisy or the reference is ambiguous.

It is also important not to mix different kinds of failure. A university should not treat *could not be checked* the same way as *checked and weakly supported*. The first case often reflects missing access to the source. The second case points to a citation context that really deserves closer reading.

Given the subjectivity of the annotation task, a more appropriate evaluation methodology for a tool of this kind would be a user study rather than a classifier benchmark. Reviewers using the system on real documents could report whether it reduced the time needed to check citations, whether it directed attention to passages that warranted closer inspection, and whether the overall review process felt more systematic. That measures what the tool is actually designed to do.

There is also a policy risk. Automated integrity tools can create false confidence if their outputs sound more certain than the evidence allows [22]. The safest use of the present pipeline is as reviewer support. It reduces search time, records where evidence was available, and highlights passages worth reading, but it does not replace human judgment.

D. Broader relevance

Even with its limits, the system fills a real gap. Universities already spend a lot of effort on monolingual similarity checking, but multilingual citation misuse is still hard to inspect by hand. A reviewer-support tool that narrows the search space and keeps an evidence trail can still be useful even if it stays conservative in its final claims.

VII. Conclusion

This paper presented a pipeline for cross-lingual citation verification based on multilingual transformer embeddings and evidence-oriented reporting. The system links citation extraction, bibliography alignment, metadata resolution, source retrieval, passage segmentation, semantic ranking, and report generation into one workflow aimed at thesis review.

The evaluation on 1000 Slovak theses shows that the general idea works, but also shows where the main limits are. DOI coverage is often sufficient to support large-scale source resolution, and multilingual ranking can often separate stronger support from weaker cases. At the same time, the biggest bottleneck is still source access. Many references can be identified, but not fully checked, because the full text is unavailable.

The main contribution of the work is therefore not a claim of perfect automatic verification. It is the design of a practical reviewer-support tool that narrows the search space, keeps uncertainty visible, and gives the reviewer concrete evidence to inspect. Future work should focus on better threshold calibration, broader citation-style support, stronger human-in-the-loop interaction, and deeper evaluation on academic prose.

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[19] S. Haan, “Semanticcite: Citation verification with ai-powered full-text analysis and evidence-based reasoning,” 2025. [Online]. Available: https://arxiv.org/abs/2511.16198
[20] L. Turnitin. (2025) ithenticate. [Online]. Available: https://www.ithenticate.com/
[21] F. informatiky Masarykovy univerzity – Vývojový tým IS MU. (2025) Odevzdej cz. [Online]. Available: https://odevzdej.cz/
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[23] T. Sağlam and L. Schmid, “Evaluating software plagiarism detection in the age of ai: Automated obfuscation and lessons for academic integrity,” 2025. [Online]. Available: https://arxiv.org/abs/2505.20158
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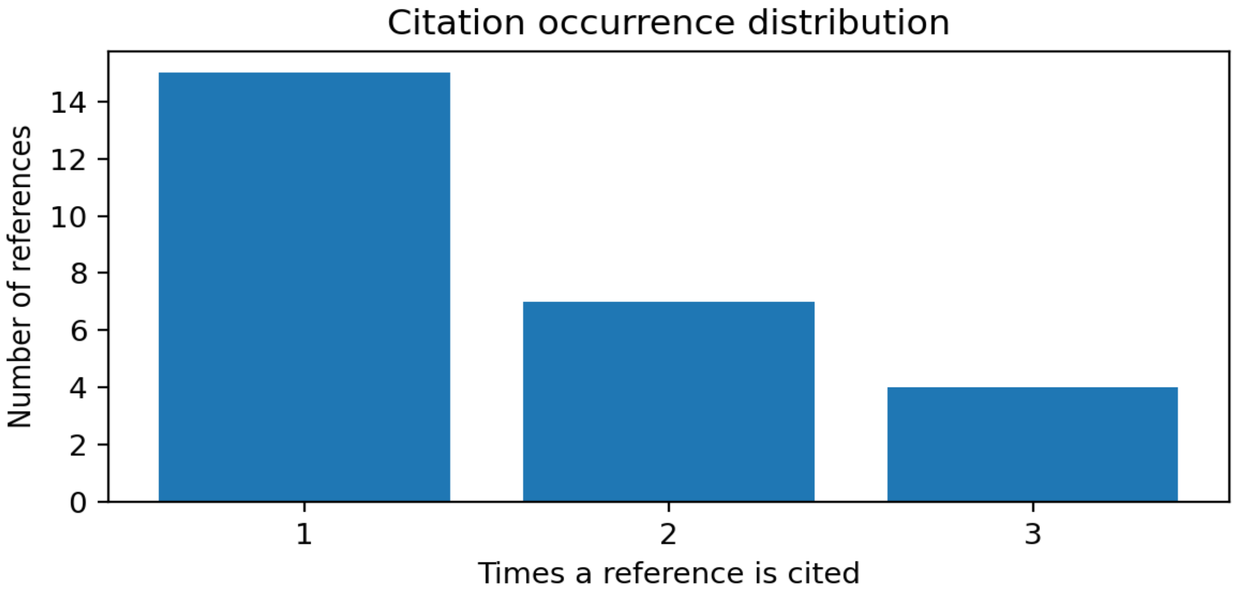
[25] “Grobid,” https://github.com/kermitt2/grobid, 2008–2026.
[26] D. Chung, “Challenges of translating idiomatic expressions: A cross-linguistic analysis at a university in hanoi, vietnam,” *International Journal of Social Science and Human Research*, vol. 07, 2024.

Citation Verification Protocol

Source PDF: D:\Uni\bp-clanok\.build\main.pdf

Paper statistics

Total references	27
Cited references	27
Unused references	0
Total citations (in-text)	42
Unmatched citation entries	0
Unmatched citation labels	0

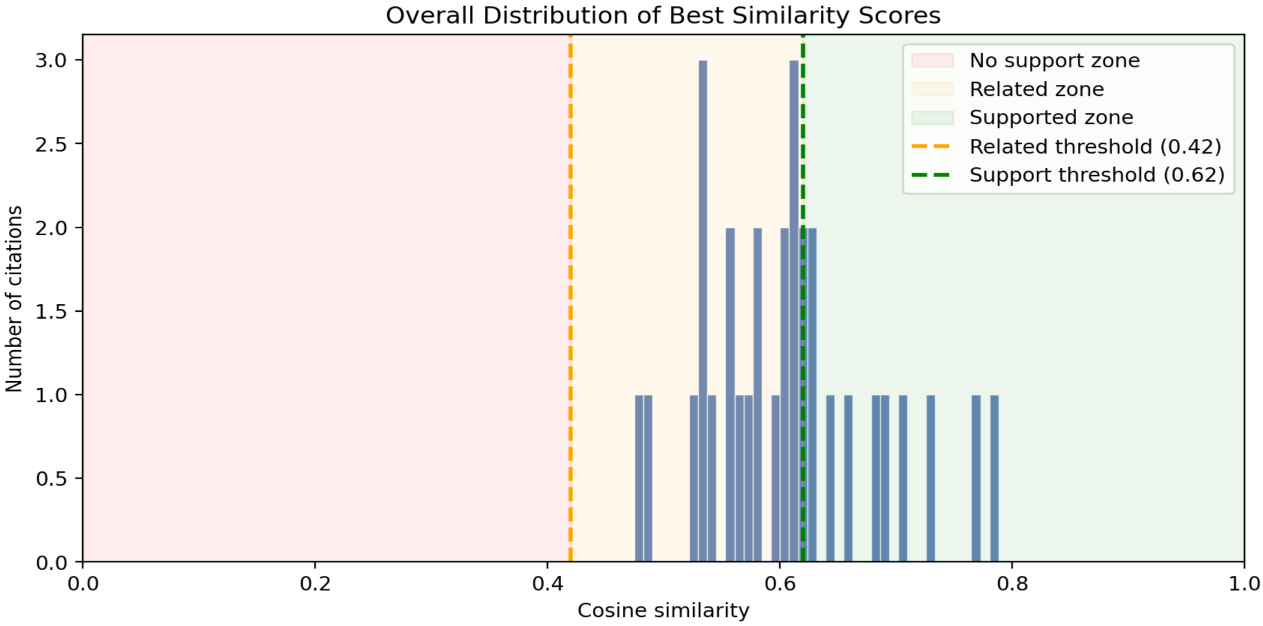


Summary

References (links): 26

Verified citing sentences: 31

Labels: supported=10, related=21, no_support=0



Reference [1]

Occurrences in work: 3

Resolved DOI: 10.1371/journal.pone.0184727

Resolved title: Accuracy of cited “facts” in medical research articles: A review of study methodology and recalculation of quotation error rate

Download status: already_indexed

Raw reference: [1] S. A. Mogull, “Accuracy of cited “facts” in medical research articles: A review of study methodology and recalculation of quotation error rate,” PLOS ONE, vol. 12, no. 9, pp. 1–17, 09 2017. [Online]. Available: https://doi.org/10.1371/journal.pone.0184727

Citing sentences and best evidence

1. Label: supported **sim:** 0.768 **hint:** body **zone:** body

Citing: A citation may be present, yet the cited source may support the claim only partially, be interpreted incorrectly, or be used in a misleading way [1] [2] [3].

- **Match 1** (sim 0.768) [body/sentence] Introduction: The primary sources cited may be misquoted, inapplicable, unreliable and occasionally even imaginary.

- **Match 2** (sim 0.692) [body/sentence] Discussion: Often, a reference listed in the work-cited list may be cited multiple times within an article and also might correspond to several different assertions (e.g., one study [26] mentioned an extreme case in which a single reference was associated with 13 in-text quotations).

- **Match 3** (sim 0.688) [body/sentence] Discussion: Alternatively, a reference may be inadvertently included in the work-cited list but not quoted in the text (as mentioned in the same study [26]).

2. Label: supported **sim:** 0.628 **hint:** unclear **zone:** abstract

Citing: That distinction matters because a citation can be formally correct and still be inaccurate, incomplete, or misleading [1] [2].

- **Match 1** (sim 0.628) [abstract/sentence] abstract: Additionally, improper secondary (or indirect) citations, which are distinguished from calculations of quotation accuracy, occur at a rate of 10.4% (3.4% to 17.5% at a 95% confidence interval).

- **Match 2** (sim 0.607) [body/sentence] Yes: Although secondary citation is procedurally improper, this measure does not provide any insight into the integrity of the information cited since the authors of the original studies did not trace the statements further and analyze for accuracy.

- **Match 3** (sim 0.596) [body/sentence] Assigning consistent definitions and criteria for quotation errors: Although source errors are procedurally improper, the quotations were not evaluated for informational or content accuracy and thus may not represent a "factual" inaccuracy.

3. Label: supported **sim:** 0.733 **hint:** body **zone:** body

Citing: Mogull [1] separates minor citation inaccuracies from major content errors, where the cited source fails to substantiate or even contradicts the assertion.

- **Match 1** (sim 0.733) [body/sentence] Discussion: g Includes secondary citations as minor content errors (the number of secondary citations was unable to be distinguished from the minor content errors).

- **Match 2** (sim 0.683) [body/sentence] Assigning consistent definitions and criteria for quotation errors: First, I defined "content errors" as an informative or "factual" inaccuracies and distinguished these errors from "source errors," which are secondary (or indirect) citations.

- **Match 3** (sim 0.677) [abstract/sentence] abstract: These content errors are predominantly, 64.8% (56.1% to 73.5% at a 95% confidence interval), major errors or cited assertions in which the referenced source either fails to substantiate, is unrelated to, or contradicts the assertion.

Reference [2]

Occurrences in work: 3

Resolved DOI: 10.1042/cs20201573

Resolved title: How accurate are citations of frequently cited papers in biomedical literature?

Download status: downloaded

Downloaded PDF: pdfs\nihms-1690179.pdf

Raw reference: [2] V. Pavlovic, T. Weissgerber, D. Stanisavljevic, T. Pekmezovic, O. Milicevic, J. M. Lazovic, A. Cirkovic, M. Savic, N. Rajovic, P. Piperac, N. Djuric, P. Madzarevic, A. Dimitrijevic, S. Randjelovic, E. Nestorovic, R. Akinyombo, A. Pavlovic, R. Ghamrawi, V. Garovic, and N. Milic, “How accurate are citations of frequently cited papers in biomedical literature?” Clin Sci (Lond), vol. 135, no. 5, pp. 671–681, mar 2021.

Citing sentences and best evidence

1. Label: related **sim:** 0.609 **hint:** weak_abstract **zone:** abstract

Citing: A citation may be present, yet the cited source may support the claim only partially, be interpreted incorrectly, or be used in a misleading way [1] [2] [3].

• **Match 1** (sim 0.609) [abstract/sentence] abstract: A citation was defined as being accurate if the cited article supported or was in accordance with the statement by citing authors.

• **Match 2** (sim 0.548) [abstract/sentence] abstract: At least one inaccurate citation was found in 11 and 15% of articles in the feasibility study and verification set, respectively, suggesting that inaccurate citations are common in biomedical literature.

• **Match 3** (sim 0.494) [abstract/sentence] abstract: -fifth of inaccurate citations were due to chains of inaccurate citations.

2. Label: related **sim:** 0.556 **hint:** weak_abstract **zone:** abstract

Citing: That distinction matters because a citation can be formally correct and still be inaccurate, incomplete, or misleading [1] [2].

• **Match 1** (sim 0.556) [abstract/sentence] abstract: -fifth of inaccurate citations were due to chains of inaccurate citations.

• **Match 2** (sim 0.508) [abstract/paragraph] abstract: Abstract Citations are an important, but often overlooked, part of every scientific paper. They allow the reader to trace the flow of evidence, serving as a gateway to relevant literature. Most scientists are aware of citations’ errors, but few appreciate the prevalence of these problems. The purpose of the present study was to examine how often frequently cited papers in biomedical scientific literature are cited inaccurately. The study include…

• **Match 3** (sim 0.505) [abstract/sentence] abstract: At least one inaccurate citation was found in 11 and 15% of articles in the feasibility study and verification set, respectively, suggesting that inaccurate citations are common in biomedical literature.

3. Label: related **sim:** 0.475 **hint:** weak_abstract **zone:** abstract

Citing: [2] show that such claim-support failures also occur in real published literature.

• **Match 1** (sim 0.475) [abstract/sentence] abstract: At least one inaccurate citation was found in 11 and 15% of articles in the feasibility study and verification set, respectively, suggesting that inaccurate citations are common in biomedical literature.

• **Match 2** (sim 0.436) [abstract/paragraph] abstract: Abstract Citations are an important, but often overlooked, part of every scientific paper. They allow the reader to trace the flow of evidence, serving as a gateway to relevant literature. Most scientists are aware of citations’ errors, but few appreciate the prevalence of these problems. The purpose of the present study was to examine how often frequently cited papers in biomedical scientific literature are cited inaccurately. The study include…

- **Match 3** (sim 0.403) [abstract/sentence] abstract: The study included an active participation of the first authors of included papers; to first-hand verify the citations accuracy.

Reference [3]

Occurrences in work: 2

Resolved DOI: 10.1177/17470161211068745

Resolved title: Scientific journals must be alert to potential manipulation in citations and referencing

Download status: already_indexed

Raw reference: [3] M. Mehregan, “Scientific journals must be alert to potential manipulation in citations and referencing,” Research Ethics, vol. 18, no. 2, pp. 163–168, 2022. [Online]. Available: https://doi.org/10.1177/17470161211068745

Citing sentences and best evidence

1. Label: related **sim:** 0.572 **hint:** unclear **zone:** body

Citing: A citation may be present, yet the cited source may support the claim only partially, be interpreted incorrectly, or be used in a misleading way [1] [2] [3].

• **Match 1** (sim 0.572) [body/sentence] Topic Piece: This may in part be attributable to unintentional errors in citation that are related to typographical mistakes and that can occur due to the negligence and carelessness of the authors; other times, however, it appears that there are many cases in which the errors in the references are due to academic misconduct (Wren and Georgescu, 2020), which is an unethical practice that involves knowingly and deliberately manipulating citations.

• **Match 2** (sim 0.558) [body/sentence] Topic Piece: In other words, citation manipulation happens when there is no connection between the cited reference and the scholarly content of the manuscript, and the contained reference does not support the information provided in the manuscript (COPE Council, 2019;National Academy of Sciences, National Academy of Engineering, and Institute of Medicine, 2009).

• **Match 3** (sim 0.538) [body/sentence] Topic Piece: Citation manipulation is falsely citing a reference that was never utilized and is only included in the manuscript with the intention of increasing the number of its citations (COPE Council, 2019).

2. Label: related **sim:** 0.578 **hint:** unclear **zone:** body

Citing: Mehregan [3] further notes that some unsupported citations may reflect deliberate manipulation rather than simple carelessness.

• **Match 1** (sim 0.578) [body/sentence] Topic Piece: This may in part be attributable to unintentional errors in citation that are related to typographical mistakes and that can occur due to the negligence and carelessness of the authors; other times, however, it appears that there are many cases in which the errors in the references are due to academic misconduct (Wren and Georgescu, 2020), which is an unethical practice that involves knowingly and deliberately manipulating citations.

• **Match 2** (sim 0.490) [body/sentence] Topic Piece: Citation manipulation is falsely citing a reference that was never utilized and is only included in the manuscript with the intention of increasing the number of its citations (COPE Council, 2019).

• **Match 3** (sim 0.482) [body/sentence] Topic Piece: In other words, citation manipulation happens when there is no connection between the cited reference and the scholarly content of the manuscript, and the contained reference does not support the information provided in the manuscript (COPE Council, 2019;National Academy of Sciences, National Academy of Engineering, and Institute of Medicine, 2009).

Reference [4]

Occurrences in work: 2

Resolved DOI: 10.1109/incacct57535.2023.10141785

Resolved title: A Review on diverse algorithms used in the context of Plagiarism Detection

Download status: already_indexed

Raw reference: [4] A. Pokharana and U. Garg, “A review on diverse algorithms used in the context of plagiarism detection,” in 2023 International Conference on Advancement in Computation & Computer Technologies (InCACCT), 2023, pp. 1–6.

Citing sentences and best evidence

1. Label: supported **sim:** 0.640 **hint:** body **zone:** body

Citing: Plagiarism is usually associated with direct copying, but in practice the broader issue also includes misleading paraphrase, weak attribution, and reuse of ideas that is formally cited but still ethically problematic [4].

- **Match 1** (sim 0.640) [body/paragraph] • Problem of Protecting and Preserving Data: Along with that if you are rewriting or rephrased the preexisting work or thoughts into your own words, then this is also compulsory to cite the work, otherwise it is also the case of plagiarism. The Latin origins plagiarius, an enslaver, and plagiarism, to steal, are where the word plagiarism comes from. "The theft of another author's work and the presentation of it as one's own, constitutes plagiarism and is a significant violation of the ethi…

- **Match 2** (sim 0.634) [body/paragraph] • Problem of Protecting and Preserving Data: Given that it is impossible to properly study the meaning of natural languages, it is exceedingly difficult to provide an exact definition of plagiarism. Use of someone else's word without giving due credit is known as plagiarism. Written content (articles, books, newspapers, magazines, emails), programming language code, photographs, painting, pictures, spoken and multimedia goods like speeches, lectures [3], YouTube videos, movies etc., music …

- **Match 3** (sim 0.628) [body/section] • Problem of Protecting and Preserving Data: In this review paper plagiarism detection issue has been picked up of detailed study III. PLAGIARISM DETECTION Plagiarism in the academia is a major and pervasive problem. Utilization of any pre -existing work or some preexisting words/statements in any form is considered at plagiarized content, according to the Oxford Dictionary. The Office of Student Judicial Affairs states that when we use another person's work without mentioning any credit i…

2. Label: related **sim:** 0.539 **hint:** unclear **zone:** body

Citing: Reviews of the field describe a gradual move from simple text matching toward more semantic and structural approaches, mainly because reused text is often changed before it appears in the suspicious document [4].

- **Match 1** (sim 0.539) [body/sentence] F. Structure-Based Methods:: This structure-based method employs context-based similarities, that means, what is the sense behind utilizing the word in the new document, as opposed to the methods mentioned above, which were designed based on lexical, syntactic, and semantic aspects of the text in texts to detect common work in various publications.

- **Match 2** (sim 0.530) [body/section] E. Fuzzy-Based Methods:: This method analyses the document similarity between the range zero to one, zero state that completely different document and no similarity between work, at the same time one means 100% matched and similar work. Each written word in these files relates to a level of similarity, therefore the texts written in the documents are represented using a set of terms with comparable meanings, and sets are regarded as fuzzy in this case. This technique is…

- **Match 3** (sim 0.530) [body/paragraph] E. Fuzzy-Based Methods:: This method analyses the document similarity between the range zero to one, zero state that completely different document and no similarity between work, at the same time one means 100% matched and similar work. Each written word in these files relates to a level of similarity, therefore the texts written in the documents are represented using a set of terms with comparable meanings, and sets are regarded as fuzzy in this case. This technique is…

Reference [5]

Occurrences in work: 1

Resolved DOI: 10.1007/s10579-009-9114-z

Resolved title: Cross-language plagiarism detection

Download status: already_indexed

Raw reference: [5] M. Potthast, A. Barr´on-Cede˜no, B. Stein, and P. Rosso, “Cross-language plagiarism detection,” Language Resources and Evaluation, vol. 45, no. 1, pp. 45–62, Mar 2011. [Online]. Available: https://doi.org/10.1007/s10579-009-9114-z

Citing sentences and best evidence

1. Label: supported **sim:** 0.687 **hint:** body **zone:** body

Citing: This places the task between cross-lingual plagiarism detection, which studies reuse across languages [5], and citation verification, which asks whether the cited source supports the citing statement [6].

- **Match 1** (sim 0.687) [body/paragraph] Introduction: The paper in hand investigates a particular kind of text plagiarism, namely the detection of plagiarism across languages, sometimes called translation plagiarism. The different kinds of text plagiarism are organized in Figure 1. Cross-language plagiarism, shown encircled, refers to cases where an author translates text from another language and then integrates the translated text into his/her own writing. It is reasonable to assume that plagiari…

- **Match 2** (sim 0.685) [body/sentence] Large part of document Document model comparison Without reference: style analysis With reference: fingerprinting: Apart from being an important practical problem, the detection of cross-language plagiarism also poses a research challenge, since the syntactical similarity between source sections and plagiarized sections found in the monolingual setting is more or less lost across languages.

- **Match 3** (sim 0.683) [body/sentence] Introduction: The paper in hand investigates a particular kind of text plagiarism, namely the detection of plagiarism across languages, sometimes called translation plagiarism.

Reference [6]

Occurrences in work: 2

Resolved DOI: 10.1021/acs.chas.4c00097.s001

Resolved title: Can Large Language Models (LLMs) Act as Virtual Safety Officers?

Download status: already_indexed

Raw reference: [6] A. Zeeb and P. Olsson, “Citation evaluation using large language models (llms) : Can llms evaluate citations in scholarly documents? an experimental study on chatgpt,” p. 43, 2024.

Citing sentences and best evidence

1. Label: related **sim:** 0.615 **hint:** unclear **zone:** body

Citing: This places the task between cross-lingual plagiarism detection, which studies reuse across languages [5], and citation verification, which asks whether the cited source supports the citing statement [6].

• **Match 1** (sim 0.615) [body/paragraph] : This broad capability makes LLMs particularly valuable for citation verification. In academic writing, ensuring that citations accurately reflect the cited sources is crucial. However, this task can be challenging due to the complexities of language and context. The ability of LLMs to analyse text could potentially enable it to assess whether a citation properly supports the claims made and whether it has been taken out of context.

• **Match 2** (sim 0.587) [body/sentence] Related work: Their research categorises citation errors into three categories: omissions, which occur when authors overlook relevant research, incorrect references, which involve errors in citing sources such as incorrect author names and wrong titles, and quotation errors, which misreport the findings of cited studies.

• **Match 3** (sim 0.584) [body/section] Implementation: Here, the implementation of this study, which aimed to use the LLMs ChatGPT 3.5 and 4 to evaluate citations in scholarly papers, will be discussed. The first step is reviewing relevant literature to construct a dataset. The aim of this dataset construction was to identify three primary types of citations: paraphrasing, direct quotation and summarising. Citations were extracted along with their corresponding source texts, giving each a boolean va…

2. Label: related **sim:** 0.608 **hint:** weak_abstract **zone:** abstract

Citing: Zeeb and Olsson explicitly study whether large language models can evaluate citations in scholarly documents [6].

• **Match 1** (sim 0.608) [abstract/sentence] abstract: This study investigates the capacity of Large Language Models (LLMs), specifically ChatGPT 3.5 and 4, to evaluate citations in scholarly papers.

• **Match 2** (sim 0.574) [body/sentence] Model Evaluation: To evaluate the models, a number of experiments is conducted, focusing on their ability to classify citations as valid or invalid.

• **Match 3** (sim 0.567) [abstract/paragraph] abstract: This study investigates the capacity of Large Language Models (LLMs), specifically ChatGPT 3.5 and 4, to evaluate citations in scholarly papers. Given the importance of accurate citations in academic writing, the goal was to determine how well these models can assist in verifying citations. A series of experiments were conducted using a dataset of our own creation. This dataset includes the three main citation categories: Direct Quotation, Parap…

Reference [7]

Occurrences in work: 2

Resolved DOI: 10.32672/jnkti.v8i1.8790.s1099

Resolved title: Turnitin

Download status: downloaded

Downloaded PDF: pdfs\10.32672_jnkti.v8i1.8790.s1099.pdf

Raw reference: [7] L. Turnitin. (2025) Turnitin. [Online]. Available: https://www.turnitin.com/

Verification status: empty_reference_index

Reference [8]

Occurrences in work: 3

Download status: no_valid_doi

Raw reference: [8] C. SR. (2025) Centr´alny register z´avere■cn´ych a kvalifika■cn´ych pr´ac. [Online]. Available: https://crzp.cvtisr.sk/

Verification status: no_source_text

Reference [9]

Occurrences in work: 1

Resolved DOI: 10.1177/0165551518787696

Resolved title: Cross-lingual text alignment for fine-grained plagiarism detection

Download status: already_indexed

Raw reference: [9] N. Ehsan, A. Shakery, and F. W. Tompa, “Cross-lingual text alignment for fine-grained plagiarism detection,” J. Inf. Sci., vol. 45, no. 4, p. 443–459, Aug. 2019. [Online]. Available: https://doi.org/10.1177/0165551518787696

Citing sentences and best evidence

1. Label: supported **sim:** 0.657 **hint:** mostly_abstract **zone:** abstract

Citing: Early approaches relied on dictionary-based [9] translation or syntax-based [10] comparison.

- **Match 1** (sim 0.657) [abstract/sentence] abstract: In both steps, our approach uses a dictionary to obtain translations of individual terms instead of using a machine translation system to convert longer passages from one language to another.
- **Match 2** (sim 0.493) [abstract/sentence] abstract: We used a weighting scheme to distinct multiple translations of the terms.
- **Match 3** (sim 0.476) [abstract/sentence] abstract: Fast and easy access to a wide range of documents in various languages, in conjunction with the wide availability of translation and editing tools, has led to the need to develop effective tools for detecting cross-lingual plagiarism.

Reference [10]

Occurrences in work: 1

Resolved DOI: 10.1016/j.eswa.2017.07.006

Resolved title: Text plagiarism classification using syntax based linguistic features

Download status: already_indexed

Raw reference: [10] V. K and D. Gupta, “Text plagiarism classification using syntax based linguistic features,” Expert Systems with Applications, vol. 88, pp. 448–464, 2017. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S095741741730475X

Citing sentences and best evidence

1. Label: related **sim:** 0.532 **hint:** unclear **zone:** body

Citing: Early approaches relied on dictionary-based [9] translation or syntax-based [10] comparison.

• **Match 1** (sim 0.532) [body/sentence] Existing systems: (2010) utilized features which included the similarity scores computed from the N-gram representations, language models, longest common subsequence, dependency relations and using basic pre-processing techniques such as stop word removal, number replacements etc.

• **Match 2** (sim 0.521) [body/sentence] Comparisons with the baseline systems: This approach used features based on different similarity scores.

• **Match 3** (sim 0.515) [body/sentence] Introduction: • An approach that utilizes minimal and effective syntax based linguistic features for plagiarism classification.

Reference [11]

Occurrences in work: 1

Resolved DOI: 10.1109/conmedia46929.2019.8981821

Resolved title: String Matching based Plagiarism Detection for Document in Bahasa Indonesia

Download status: downloaded

Downloaded PDF:
pdfs\String_Matching_based_Plagiarism_Detection_for_Document_in_Bahasa_Indonesia.pdf

Raw reference: [11] W. G. S. Parwita, I. G. A. A. D. Indradewi, and I. N. S. W. Wijaya, “String matching based plagiarism detection for document in bahasa indonesia,” in 2019 5th International Conference on New Media Studies (CONMEDIA), 2019, pp. 54–58.

Verification status: empty_reference_index

Reference [12]

Occurrences in work: 1

Resolved DOI: 10.3115/v1/d14-1153

Resolved title: Intrinsic Plagiarism Detection using N-gram Classes

Download status: already_indexed

Raw reference: [12] I. Bensalem, P. Rosso, and S. Chikhi, “Intrinsic plagiarism detection using n-gram classes,” in Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), A. Moschitti, B. Pang, and W. Daelemans, Eds. Doha, Qatar: Association for Computational Linguistics, Oct. 2014, pp. 1459–1464. [Online]. Available: https://aclanthology.org/D14-1153/

Citing sentences and best evidence

1. Label: related **sim:** 0.528 **hint:** unclear **zone:** body

Citing: Classical lexical methods such as string matching [11] and n-gram [12] comparison can still work when the suspicious text stays close to the source.

- **Match 1** (sim 0.528) [body/paragraph] Introduction and Related Works: Unlike the authorship attribution and verification, where the examined text and the potential author text are separate (and hence their writing styles could be readily characterized and compared), these two parts are both merged in the same document with unknown boundaries. Furthermore, the plagiarized fragments in a suspicious document might stem from different authors, which renders the computational characterization of plagiarism difficult.

- **Match 2** (sim 0.503) [body/paragraph] Introduction and Related Works: As opposed to the problem of authorship clustering, where the task is merely to attribute already defined fragments of a given document to different authors, the segmentation is a crucial and inevitable task in a real scenario of intrinsic plagiarism detection. Indeed, a granular segmentation may lead to an undependable style analysis, and a coarse segmentation may prevent the identification of the short plagiarized texts.

- **Match 3** (sim 0.493) [body/sentence] Introduction and Related Works: Indeed, a granular segmentation may lead to an undependable style analysis, and a coarse segmentation may prevent the identification of the short plagiarized texts.

Reference [13]

Occurrences in work: 2

Resolved DOI: 10.3389/fcomp.2025.1504725

Resolved title: Plagiarism types and detection methods: a systematic survey of algorithms in text analysis

Download status: already_indexed

Raw reference: [13] A. Amirzhanov, C. Turan, and A. Makhmutova, “Plagiarism types and detection methods: a systematic survey of algorithms in text analysis,” Frontiers in Computer Science, vol. Volume 7 - 2025, 2025. [Online]. Available: https://www.frontiersin.org/journals/computer-science/articles/10.3389/fcomp. 2025.1504725

Citing sentences and best evidence

1. Label: related **sim:** 0.487 **hint:** unclear **zone:** body

Citing: Their weakness is that performance drops when the author translates, paraphrases, or compresses the original meaning more aggressively [13].

- **Match 1** (sim 0.487) [body/sentence] . Types of plagiarism: • Translation plagiarism: translating content from one language to another without citation.
- **Match 2** (sim 0.470) [body/sentence] . Types of plagiarism: • Paraphrased plagiarism: rewriting the original text while retaining the core meaning.
- **Match 3** (sim 0.445) [body/sentence] . Semantic similarity-based approaches: Semantic similarity methods detect plagiarism by analyzing the meaning behind words, going beyond surface-level text comparison.

2. Label: supported **sim:** 0.788 **hint:** body **zone:** body

Citing: [13] describe cross-lingual plagiarism detection as a problem that increasingly requires multilingual rather than purely lexical comparison.

- **Match 1** (sim 0.788) [body/sentence] Evolution of detection methodologies in reviewed papers: -A growing need for cross-language plagiarism detection.
- **Match 2** (sim 0.776) [body/sentence] Paper selection methodology: -Monolingual plagiarism detection vs. cross-lingual plagiarism detection methods.
- **Match 3** (sim 0.758) [body/sentence] Cross-language and multilingual approaches: Cross-language plagiarism detection methods address the challenge of identifying plagiarism in translated texts.

Reference [14]

Occurrences in work: 1

Resolved DOI: 10.18653/v1/N19-1423

Resolved title: Spectrum-BERT: Pre-training of Deep Bidirectional Transformers for Spectral Classification of Chinese Liquors_supp1-3374300.docx

Download status: downloaded

Downloaded PDF: pdfs\N19-1423.pdf

Raw reference: [14] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," Minneapolis, Minnesota, pp. 4171–4186, Jun. 2019. [Online]. Available: https://aclanthology.org/N19-1423/

Citing sentences and best evidence

1. Label: related **sim:** 0.536 **hint:** unclear **zone:** body

Citing: BERT [14] showed that contextual pretraining can learn richer text representations than simple lexical features.

• **Match 1** (sim 0.536) [body/paragraph] Introduction: • We demonstrate the importance of bidirectional pre-training for language representations. Unlike Radford et al. (2018), which uses unidirectional language models for pre-training, BERT uses masked language models to enable pretrained deep bidirectional representations. This is also in contrast to Peters et al. (2018a), which uses a shallow concatenation of independently trained left-to-right and right-to-left LMs.

• **Match 2** (sim 0.532) [body/section] Pre-training data: The pre-training procedure largely follows the existing literature on language model pre-training. For the pre-training corpus we use the BooksCorpus (800M words) (Zhu et al., 2015) and English Wikipedia (2,500M words). For Wikipedia we extract only the text passages and ignore lists, tables, and headers. It is critical to use a document-level corpus rather than a shuffled sentence-level corpus such as the Billion Word Benchmark (Chelba et al., …

• **Match 3** (sim 0.529) [body/sentence] A.4 Comparison of BERT, ELMo ,and: BERT is trained on the BooksCorpus (800M words) and Wikipedia (2,500M words).

Reference [15]

Occurrences in work: 2

Resolved DOI: 10.18653/v1/2020.acl-main.747

Resolved title: Unsupervised Cross-lingual Representation Learning at Scale

Download status: already_indexed

Raw reference: [15] A. Conneau, K. Khandelwal, N. Goyal, V. Chaudhary, G. Wenzek, F. Guzm´an, E. Grave, M. Ott, L. Zettlemoyer, and V. Stoyanov, “Unsupervised cross-lingual representation learning at scale,” in Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, D. Jurafsky, J. Chai, N. Schluter, and J. Tetreault, Eds. Online: Association for Computational Linguistics, Jul. 2020, pp. 8440–8451. [Online]. Available: https://aclanthology.org/2020.acl-main.747/

Citing sentences and best evidence

1. Label: supported **sim:** 0.626 **hint:** body **zone:** body

Citing: XLM-R [15] extended this idea to many languages and remains a strong baseline for cross-lingual semantic tasks.

- **Match 1** (sim 0.626) [body/sentence] Analysis and Results: We then present the results of XLM-R on cross-lingual understanding and GLUE.
- **Match 2** (sim 0.614) [body/paragraph] Introduction: Multilingual masked language models (MLM) like mBERT (Devlin et al., 2018) and XLM (Lample and Conneau, 2019) have pushed the stateof-the-art on cross-lingual understanding tasks by jointly pretraining large Transformer models (Vaswani et al., 2017) on many languages. These models allow for effective cross-lingual transfer, as seen in a number of benchmarks including cross-lingual natural language inference (Bowman et al., 2015;Williams et al., …
- **Match 3** (sim 0.599) [body/sentence] Model: While XLM-R is among the largest models partly due to its large embedding layer, it has a similar number of parameters than XLM-100, and remains significantly smaller that recently introduced Transformer models for multilingual MT and transfer learning.

2. Label: supported **sim:** 0.709 **hint:** mostly_abstract **zone:** abstract

Citing: The testing used an XLM-R-based [15] sentence embedding model because it offers broad language coverage and good practical performance in cross-lingual similarity tasks [17].

- **Match 1** (sim 0.709) [abstract/paragraph] abstract: This paper shows that pretraining multilingual language models at scale leads to significant performance gains for a wide range of crosslingual transfer tasks. We train a Transformerbased masked language model on one hundred languages, using more than two terabytes of filtered CommonCrawl data. Our model, dubbed XLM-R, significantly outperforms multilingual BERT (mBERT) on a variety of cross-lingual benchmarks, including +14.6% average accuracy …
- **Match 2** (sim 0.660) [body/sentence] Introduction: We present XLM-R a transformer-based multilingual masked language model pre-trained on text in 100 languages, which obtains state-of-the-art performance on cross-lingual classification, sequence labeling and question answering.
- **Match 3** (sim 0.656) [abstract/sentence] abstract: Finally, we show, for the first time, the possibility of multilingual modeling without sacrificing perlanguage performance; XLM-R is very competitive with strong monolingual models on the GLUE and XNLI benchmarks.

Reference [16]

Occurrences in work: 1

Resolved DOI: 10.18653/v1/2023.emnlp-main.813

Resolved title: XLM-V: Overcoming the Vocabulary Bottleneck in Multilingual Masked Language Models

Download status: already_indexed

Raw reference: [16] D. Liang, H. Gonen, Y. Mao, R. Hou, N. Goyal, M. Ghazvininejad, L. Zettlemoyer, and M. Khabsa, “XLM-V: Overcoming the vocabulary bottleneck in multilingual masked language models,” Singapore, pp. 13 142–13 152, Dec. 2023. [Online]. Available: https://aclanthology.org/2023.emnlp-main.813/

Citing sentences and best evidence

1. Label: related **sim:** 0.602 **hint:** unclear **zone:** body

Citing: Newer encoder families such as XLM-V [16] continue this line by improving multilingual coverage and representation quality.

- **Match 1** (sim 0.602) [body/sentence] Fully trained model: Table 3: XLM-V outperforms XLM-R on cross-lingual transfer on every language in XNLI with outsized improvements on the lower-resource languages, Swahili and Urdu.
- **Match 2** (sim 0.534) [abstract/sentence] abstract: This vocabulary bottleneck limits the representational capabilities of multilingual models like XLM-R.
- **Match 3** (sim 0.531) [body/sentence] Sentencepiece: This method is used by both XLM-R and our work. et al. (2020) proposed an approach to multilingual vocabulary construction that balances the trade-off between optimizing for cross-lingual subword sharing and the need for robust representation of individual languages.

Reference [17]

Occurrences in work: 2

Resolved DOI: 10.1007/s10579-024-09759-3

Resolved title: PESTS: Persian_English cross lingual corpus for semantic textual similarity

Download status: already_indexed

Raw reference: [17] M. Abdous, P. Piroozfar, and B. MinaeiBidgoli, “Pests: Persian english cross lingual corpus for semantic textual similarity,” Language Resources and Evaluation, vol. 59, no. 2, pp. 1179–1199, Jun 2025. [Online]. Available: https://doi.org/10.1007/s10579-024-09759-3

Citing sentences and best evidence

1. Label: related **sim:** 0.568 **hint:** unclear **zone:** body

Citing: Recent studies report good results for transformer-based multilingual embeddings in paraphrase-oriented and semantic cross-lingual settings [17].

• **Match 1** (sim 0.568) [body/sentence] Conclusion: Semantic textual similarity represents a significant area of research in natural language processing, particularly in the context of cross-lingual applications.

• **Match 2** (sim 0.546) [body/paragraph] Related works: Ferrero et al. (Ferrero et al., 2016) provided a dataset to assess the similarity of cross lingual texts that can be very useful in fraud detection. The dataset provided is multilingual, containing texts in French, English, and Spanish languages. It facilitates cross-lingual alignment of information at different levels, such as documents, sentences, and fragments. The dataset comprises both human-generated texts and texts obtained through machin…

• **Match 3** (sim 0.543) [body/sentence] Results and discussion: Some of these Transformers are multilingual, such as Multilingual BERT (mBERT) and XLM (Cross-lingual Language Model).

2. Label: related **sim:** 0.606 **hint:** unclear **zone:** body

Citing: The testing used an XLM-R-based [15] sentence embedding model because it offers broad language coverage and good practical performance in cross-lingual similarity tasks [17].

• **Match 1** (sim 0.606) [body/sentence] Experiments: To assess the performance of the developed model, we conducted tests and evaluations using cross-lingual semantic textual similarity test data.

• **Match 2** (sim 0.586) [body/section] Results and discussion: In this paper, we have performed experiments on the generated corpus, employing a range of Transformer-based language models. Our objective was to assess and compare the performance of these models in tasks associated with quantifying semantic textual similarity. Transformers have been developed to address the sequencing problem in neural networks. They demonstrate exceptional proficiency in handling sequences of data, such as the words in a sen…

• **Match 3** (sim 0.586) [body/paragraph] Results and discussion: In this paper, we have performed experiments on the generated corpus, employing a range of Transformer-based language models. Our objective was to assess and compare the performance of these models in tasks associated with quantifying semantic textual similarity. Transformers have been developed to address the sequencing problem in neural networks. They demonstrate exceptional proficiency in handling sequences of data, such as the words in a sen…

Reference [18]

Occurrences in work: 1

Resolved DOI: 10.18653/v1/n19-1361

Resolved title: Structural Scaffolds for Citation Intent Classification in Scientific Publications

Download status: downloaded

Downloaded PDF: pdfs\10.18653_v1_n19-1361.pdf

Raw reference: [18] A. Cohan, W. Ammar, M. van Zuylen, and F. Cady, "Structural scaffolds for citation intent classification in scientific publications," in Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), J. Burstein, C. Doran, and T. Solorio, Eds. Minneapolis, Minnesota: Association for Computational Linguistics, Jun. 2019, pp. 3586–3596. [Online]. Available: https://aclanthology.…

Citing sentences and best evidence

1. Label: related **sim:** 0.581 **hint:** unclear **zone:** body

Citing: SciCite, for example, classifies citation intent in scientific writing [18].

• **Match 1** (sim 0.581) [body/sentence] Structural scaffolds: In scientific writing there is a connection between the structure of scientific papers and the intent of citations.

• **Match 2** (sim 0.538) [body/section] Introduction: Citations play a unique role in scientific discourse and are crucial for understanding and analyzing scientific work (Luukkonen, 1992;Leydesdorff, 1998). They are also typically used as the main measure for assessing impact of scientific publications, venues, and researchers (Li and Ho, 2008). The nature of citations can be different. Some citations indicate direct use of a method while some others merely serve as acknowledging a prior work. The…

• **Match 3** (sim 0.538) [body/paragraph] Introduction: Citations play a unique role in scientific discourse and are crucial for understanding and analyzing scientific work (Luukkonen, 1992;Leydesdorff, 1998). They are also typically used as the main measure for assessing impact of scientific publications, venues, and researchers (Li and Ho, 2008). The nature of citations can be different. Some citations indicate direct use of a method while some others merely serve as acknowledging a prior work. The…

Reference [19]

Occurrences in work: 1

Resolved DOI: 10.48550/arxiv.2511.16198

Resolved title: SemanticCite: Citation Verification with AI-Powered Full-Text Analysis and Evidence-Based Reasoning

Download status: already_indexed

Raw reference: [19] S. Haan, “Semanticcite: Citation verification with ai-powered full-text analysis and evidence-based reasoning,” 2025. [Online]. Available: https://arxiv.org/abs/2511.16198

Citing sentences and best evidence

1. Label: related **sim:** 0.619 **hint:** unclear **zone:** body

Citing: Haan’s SemanticCite is even closer at the systems level because it approaches citation verification as a workflow over full texts, citation contexts, and evidence passages [19].

• **Match 1** (sim 0.619) [body/section] Contributions: This paper introduces SemanticCite, the first comprehensive AI-powered framework for automated fulltext citation verification. Unlike existing approaches that rely on abstract-level analysis or binary classification schemes, our system performs deep semantic analysis of complete source documents using a 4-class taxonomy that captures the nuanced relationships between citations and their references. Seman-ticCite bridges the gap between automated…

• **Match 2** (sim 0.619) [body/paragraph] Contributions: This paper introduces SemanticCite, the first comprehensive AI-powered framework for automated fulltext citation verification. Unlike existing approaches that rely on abstract-level analysis or binary classification schemes, our system performs deep semantic analysis of complete source documents using a 4-class taxonomy that captures the nuanced relationships between citations and their references. Seman-ticCite bridges the gap between automated…

• **Match 3** (sim 0.586) [body/paragraph] Conclusion: The framework's practical implications extend beyond academic manuscript review. By providing actionable guidance through detailed reasoning and evidence extraction, SemanticCite bridges the gap between automated analysis and human editorial decision-making, supporting iterative manuscript improvement rather than binary acceptance/rejection outcomes. The same 4-class taxonomy applies directly to AI-generated content verification, addressing crit…

Reference [20]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [20] L. Turnitin. (2025) ithenticate. [Online]. Available: https://www.ithenticate.com/

Verification status: no_source_text

Reference [21]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [21] F. informatiky Masarykovy univerzity – V´yvojov´y t´ym IS MU. (2025) Odevzdej cz. [Online]. Available: https://odevzdej.cz/

Verification status: no_source_text

Reference [22]

Occurrences in work: 1

Resolved DOI: 10.1080/0361526x.2024.2433256

Resolved title: The Problem with False Positives: AI Detection Unfairly Accuses Scholars of AI Plagiarism

Download status: already_indexed

Raw reference: [22] L. Giray, “The problem with false positives: Ai detection unfairly accuses scholars of ai plagiarism,” The Serials Librarian, vol. 85, no. 5-6, pp. 181–189, 2024. [Online]. Available: https://doi.org/10.1080/0361526X.2024.2433256

Citing sentences and best evidence

1. Label: related **sim:** 0.532 **hint:** unclear **zone:** body

Citing: Threshold selection, class imbalance, and overconfident interpretation can make reported performance look cleaner than the underlying evidence really is [22] [23].

• **Match 1** (sim 0.532) [body/sentence] Voices of scholars from different parts of the globe: The lack of clear rules for dealing with false positives makes things even trickier.

• **Match 2** (sim 0.528) [body/sentence] Prologue: But these tools are not perfect and rely on patterns and statistical analyses, which often fail to distinguish between legitimate academic writing and potential violations.

• **Match 3** (sim 0.486) [body/section] Algorithmic biases: One of the biggest issues with AI detection tools is the presence of algorithmic biases. These biases, often baked into the algorithms, can unfairly target specific writing styles or even cultural differences in language use. For instance, a detection tool might be more likely to flag work written by non-native English speakers, simply because their sentence structures differ from standard English conventions. This bias mirrors what Steele and A…

2. Label: related **sim:** 0.619 **hint:** unclear **zone:** body

Citing: This matters because automated integrity tools can easily look more certain than they really are [22].

• **Match 1** (sim 0.619) [body/sentence] Voices of scholars from different parts of the globe: 17 While these tools are supposed to help maintain academic integrity, they're not always perfect.

• **Match 2** (sim 0.589) [body/sentence] Healthy skepticism: As these tools become more integrated into academic and professional settings, careful consideration of their accuracy is essential.

• **Match 3** (sim 0.572) [body/sentence] Voices of scholars from different parts of the globe: As AI continues to evolve, there's a growing need for more accurate and reliable detection tools.

3. Label: related **sim:** 0.599 **hint:** unclear **zone:** abstract

Citing: Automated integrity tools can create false confidence if their outputs sound more certain than the evidence allows [22].

• **Match 1** (sim 0.599) [abstract/sentence] abstract: While AI detection tools have value, a healthy skepticism is needed due to the risk of false positives and other limitations.

• **Match 2** (sim 0.571) [body/sentence] Voices of scholars from different parts of the globe: 17 While these tools are supposed to help maintain academic integrity, they're not always perfect.

• **Match 3** (sim 0.567) [body/sentence] Voices of scholars from different parts of the globe: One of the biggest problems with AI detection tools is that they tend to produce false positives.

Reference [23]

Occurrences in work: 3

Resolved DOI: 10.48550/arxiv.2505.20158

Resolved title: Evaluating Software Plagiarism Detection in the Age of AI: Automated Obfuscation and Lessons for Academic Integrity

Download status: already_indexed

Raw reference: [23] T. Sa■glam and L. Schmid, “Evaluating software plagiarism detection in the age of ai: Automated obfuscation and lessons for academic integrity,” 2025. [Online]. Available: https://arxiv.org/abs/2505.20158

Citing sentences and best evidence

1. Label: related **sim:** 0.560 **hint:** unclear **zone:** body

Citing: Threshold selection, class imbalance, and overconfident interpretation can make reported performance look cleaner than the underlying evidence really is [22] [23].

- **Match 1** (sim 0.560) [body/sentence] Similarity Metrics: A threshold-based evaluation reduces plagiarism detection to a binary classification problem, which is insufficient due to the mentioned problem of overlap.

- **Match 2** (sim 0.552) [body/paragraph] Similarity Metrics: A common anti-pattern in existing works is evaluating plagiarism detectors using a fixed similarity threshold where scores above count as successful detec-tions, and those below do not. While this simplifies deriving precision, recall, and F1 scores, this approach is fundamentally flawed, as the threshold can arbitrarily influence results and it can be tuned to favor one approach. Since no universal threshold fits all datasets, thresholds are ch…

- **Match 3** (sim 0.523) [body/paragraph] Similarity Metrics: For these reasons, we focus on the difference in similarity between plagiarism and non-plagiarism pairs. The larger this difference, the easier it is to detect plagiarism effectively. Thus, we measure to what extent a detection approach can produce such a difference between these pairs. This avoids overabstracting the problem into a binary classification. Varying statistical measures can be used to calculate the differences between these types o…

Reference [24]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [24] Crossref. (2026) Crossref. [Online]. Available: https://www.crossref.org/

Verification status: no_source_text

Reference [25]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [25] “Grobid,” https://github.com/kermitt2/grobid, 2008–2026.

Verification status: no_source_text

Reference [26]

Occurrences in work: 1

Resolved DOI: 10.47191/ijsshr/v7-i10-39

Resolved title: Challenges of Translating Idiomatic Expressions: A Cross-Linguistic Analysis at a University in Hanoi, Vietnam

Download status: already_indexed

Raw reference: [26] D. Chung, “Challenges of translating idiomatic expressions: A cross-linguistic analysis at a university in hanoi, vietnam,” International Journal of Social Science and Human Research, vol. 07, 2024.

Citing sentences and best evidence

1. Label: supported **sim:** 0.681 **hint:** body **zone:** body

Citing: Translation adds another difficulty because idioms and terminology shifts can lower similarity even when the source is used correctly [26].

- **Match 1** (sim 0.681) [body/sentence] Semantic Issues: Semantic errors often arise when translators attempt a word-for-word translation, as the figurative meaning is entirely lost in the process.
- **Match 2** (sim 0.668) [body/sentence] I. INTRODUCTION 1.1.: Moreover, the non-literal nature of idioms often leads to misinterpretation when translators apply word-for-word translation methods.
- **Match 3** (sim 0.652) [body/sentence] 2.2.5.: This structural discrepancy can obscure the intended meaning if translators attempt a literal translation.